

HW1 Answers

(a)

$$\begin{aligned}A_v &\equiv \frac{v_{out}}{v_{in}} = GR_2 \\R_{in} &\equiv \frac{v_{in}}{i_{in}} = \frac{R_1}{1 + GR_1} \\R_{out} &= R_2\end{aligned}$$

(b)

$$\begin{aligned}A_v &= -\frac{GR_3}{1 + GR_2} \\R_{in} &= R_1 \\R_{out} &= R_3\end{aligned}$$

(c)

$$\begin{aligned}A_v &= \frac{GR_2}{1 + GR_2} \\R_{in} &= R_1 \\R_{out} &= \frac{R_2}{1 + GR_2}\end{aligned}$$

(d)

$$\begin{aligned}A_v &= \frac{(1 + K)R_2}{R_1 + (1 + K)R_2} \\R_{in} &= R_1 + (1 + K)R_2 \\R_{out} &= \frac{R_1R_2}{R_1 + (1 + K)R_2}\end{aligned}$$

(e)

$$\begin{aligned}A_v &= -K \\R_{in} &= \frac{R_1R_2}{R_2 + (1 + K)R_1} \\R_{out} &= R_3\end{aligned}$$

(f)

$$\begin{aligned}
A_v &= \frac{R_3(1 + GR_2)}{R_1 + R_2 + R_3 + GR_2R_3} \\
R_{in} &= R_1 + R_2 + R_3 + GR_2R_3 \\
R_{out} &= \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3 + GR_2R_3}
\end{aligned}$$

(g)

$$\begin{aligned}
A_v &= -\frac{GR_2R_4}{R_1 + R_2 + R_3 + GR_2R_3} \\
R_{in} &= R_1 + R_2 + R_3 + GR_2R_3 \\
R_{out} &= R_4
\end{aligned}$$

(h)

$$\begin{aligned}
A_v &= \frac{GR_1R_4}{R_1 + R_2 + R_3} \\
R_{in} &= \frac{R_1 + R_2 + R_3}{1 + GR_1} \\
R_{out} &= R_4
\end{aligned}$$

(i)

$$\begin{aligned}
A_v &= \frac{Z}{R_1} \\
R_{in} &= R_1 \\
R_{out} &= R_2 \left(1 + \frac{KZ}{R_1} \right) = \frac{R_2(R_1 + KZ)}{R_1}
\end{aligned}$$

(j)

$$\begin{aligned}
A_v &= \frac{R_3}{R_3 + (1 - K)R_2} \\
R_{in} &= \frac{R_1(R_3 + (1 - K)R_2)}{R_1 + R_3 + (1 - K)R_2} \\
R_{out} &= \frac{R_2R_3}{R_3 + (1 - K)R_2}
\end{aligned}$$